

AIKEN, SC, USA

WMO: 720601

Lat: 33.650N

Lon: 81.683W

Elev: 529

StdP: 14.42

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00193

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	27.3	7.3	8.2	27.3	12.4	10.5	34.9	21.3	51.0	19.8	53.2	3.3	40	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	95.3	76.3	93.0	75.8	90.8	75.6	82.6	90.3	80.9	86.7	78.9	84.7	8.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	164.2	86.8	79.1	154.2	84.6	77.2	144.6	81.7	46.9	89.8	44.8	86.0	42.9	83.4	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	16.2	13.3	17.4	98.6	3.6	3.0	14.8	100.8	12.7	102.5	10.6	104.2	8.0	106.4
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	43.8	48.9	54.4	63.2	71.5	78.4	80.7	79.1	74.8	64.0	52.3	48.0
	DBStd	14.76	9.92	9.34	9.11	7.23	5.93	4.08	3.27	3.39	4.99	7.40	7.42	8.99
	HDD50	646	248	126	61	3	0	0	0	0	0	2	57	149
	HDD65	2619	659	455	345	118	19	0	0	3	107	384	529	
	CDD50	5514	56	95	196	399	666	853	952	903	744	435	127	88
	CDD65	2014	2	4	16	63	220	403	487	438	297	76	4	4
	CDH74	18748	7	26	150	680	1966	3962	4805	3950	2511	653	32	6
	CDH80	7940	0	0	15	146	711	1858	2304	1740	1016	149	1	0
Wind	WSAvg	5.6	6.0	6.4	6.7	6.7	5.7	5.2	5.3	4.6	4.9	5.2	5.0	5.6
Precipitation	PrecAvg	52.2	5.1	4.5	5.4	3.7	4.0	5.4	5.4	5.3	4.0	3.0	3.0	3.7
	PrecMax	77.2	10.4	8.8	14.4	13.4	8.0	13.8	11.7	11.5	8.5	12.7	11.5	7.6
	PrecMin	35.3	1.1	1.1	1.1	0.7	0.4	0.5	1.4	0.8	0.1	0.0	0.2	0.8
	PrecStd	8.8	2.4	2.0	2.9	2.6	2.0	2.6	2.6	2.7	2.4	2.6	2.1	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	78.6	81.9	87.7	94.6	97.4	99.0	96.9	94.6	88.4	78.7	73.2
		MCWB	64.5	65.0	64.5	70.5	72.1	76.5	80.5	77.8	73.7	72.6	66.5	66.8
	2%	DB	68.2	72.5	78.9	83.8	90.2	95.0	95.5	93.1	91.0	83.7	72.8	70.3
		MCWB	61.0	62.4	63.2	69.2	72.5	76.2	78.0	75.2	73.6	70.8	62.7	64.9
	5%	DB	63.8	69.6	73.4	81.3	87.7	91.3	92.9	90.6	89.7	81.0	70.2	66.3
		MCWB	58.2	59.4	60.6	66.6	71.3	75.3	77.0	76.0	72.7	68.3	61.6	62.6
	10%	DB	60.8	63.9	70.4	78.7	83.8	90.0	90.6	89.6	86.3	77.4	66.5	63.2
		MCWB	55.1	57.2	59.2	64.4	70.4	74.8	76.9	76.0	72.3	66.4	60.3	59.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.9	67.1	68.1	75.4	79.1	83.1	84.4	84.2	79.2	77.1	69.4	69.7
		MCDB	72.0	73.5	76.9	82.4	86.0	93.3	94.8	91.4	84.7	79.9	71.0	71.9
	2%	WB	63.6	64.8	66.2	72.1	76.6	81.2	82.9	82.1	77.6	74.2	66.8	67.1
		MCDB	65.7	70.4	75.0	79.5	84.4	89.6	91.5	87.1	85.2	79.3	71.3	69.3
	5%	WB	60.3	62.7	64.0	69.6	74.0	78.9	81.3	79.9	76.0	72.5	64.1	63.6
		MCDB	63.4	66.7	70.2	76.0	81.1	86.2	87.3	83.7	83.7	77.5	67.3	64.4
	10%	WB	55.4	59.7	61.8	67.5	72.2	76.7	79.0	77.9	74.3	69.9	61.7	60.6
		MCDB	58.6	63.5	67.6	75.1	79.7	84.8	84.9	83.9	81.5	74.6	65.2	63.3
Mean Daily Temperature Range	5% DB	MDBR	20.0	20.9	22.8	23.8	21.4	20.7	20.1	19.0	20.4	22.6	22.5	18.4
		MCDBR	23.5	24.5	26.6	26.3	25.6	23.8	23.0	21.8	24.5	24.5	25.6	21.2
		MCWBR	16.1	14.8	13.6	12.2	9.5	8.0	7.6	7.4	8.2	11.6	16.2	14.9
	5% WB	MCDBR	18.8	21.1	22.7	23.4	21.2	22.2	21.6	19.0	20.9	19.4	20.4	15.8
		MCWBR	15.0	14.1	12.8	13.4	10.3	10.1	9.8	9.3	9.1	10.0	15.3	12.8
Clear-Sky Solar Irradiance	taub	0.320	0.327	0.364	0.401	0.465	0.499	0.544	0.525	0.435	0.395	0.348	0.317	
	taud	2.496	2.474	2.398	2.287	2.153	2.102	1.997	2.039	2.276	2.354	2.447	2.538	
	Ebn at Noon	284	293	289	282	264	253	241	244	263	265	270	279	
	Edh at Noon	27	31	36	42	49	51	56	53	40	34	28	25	
All-Sky Solar Radiation	RadAvg	889	1086	1429	1798	1954	2016	1924	1757	1522	1276	974	770	
	RadStd	47	113	103	128	148	141	134	87	131	150	76	73	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (57 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+331	+232

Nomenclature: See separate page